WSABI Summer Research Lab **Summer 2026**

Lab 2: Multivariable Linear Regression

Instructor: Jonathan Pipping-Gamón *Author: Ryan Brill*

2.1 Four Factors in Basketball

Basketball has many different components that contribute to a team's success: shooting accuracy, defense, rebounding, turnovers, free throws, etc. Which of these correlates with winning? Which matters the most? Which matters the least?

2.1.1 The Work of Dean Oliver

Dean Oliver defined four factors that contribute to winning:

- **Scoring**: shooting accuracy
- **Crashing**: rebounding efficiency
- **Protecting**: turnover rates
- **Attacking**: free throw shooting

This was a big innovation because expressing these quantities in percentage or rate terms helps normalize for pace of play. A fast team and a slow team might have very different raw totals, but rates let us compare them more fairly.

2.1.2 Data

Your task will be to use multivariable regression to assess the relative impact of the four factors on winning. Let y be team wins in a season. Define the following explanatory variables so that larger values are better for the team:

$$
\begin{aligned}
x_1 &= eFG\% - opp\ eFG\% \\
x_2 &= OREB\% + DREB\% - 100 \\
x_3 &= opp\ TOV\% - TOV\% \\
x_4 &= FT\ rate - opp\ FT\ rate.
\end{aligned}
$$

Here, x_1 measures shooting advantage, x_2 measures rebounding advantage, x_3 measures turnover advantage, and x_4 measures free-throw-rate advantage.

The effective field goal percentage is defined as

$$eFG\% = \frac{FG + \frac{1}{2}3PT}{FGA},$$

which weights successful three-pointers 50% more than successful two-pointers.

Also, $TOV\%$ is turnover percentage, $OREB\%$ is offensive rebounding percentage, $DREB\%$ is defensive rebounding percentage, and $FT\ rate$ measures how often a team gets to the free throw line.

2.1.3 Your Assignment

Task 1: Get to know the data.

- Find each variable's mean, standard deviation, maximum, and minimum.
- Plot the marginal distribution of each explanatory variable.
- Make scatterplots of wins against each of the four factors.
- Find the correlation between each pair of explanatory variables.
- Which variables seem most strongly related to wins before fitting a multivariable model?

Task 2: Model the data. Fit a multivariable linear regression model of the form

$$E[y \mid x_1, x_2, x_3, x_4] = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4.$$

- Write down the fitted regression equation.
- Interpret each coefficient in context. For example, what does $\widehat{\beta}_1$ mean?
- Since all four x's were defined so that larger values are better, do the signs of the estimated coefficients make sense?
- Which factor appears most strongly associated with winning?
- Which factors appear weakly associated with winning after adjusting for the other factors?

Task 3: Standardize the predictors. Create standardized versions of the explanatory variables:

$$z_j = \frac{x_j - \bar{x}_j}{s_j}, \qquad j = 1, 2, 3, 4,$$

where $\bar{x}_j$ is the sample mean of x_j and s_j is its sample standard deviation. Then fit

$$E[y \mid z_1, z_2, z_3, z_4] = \alpha_0 + \alpha_1 z_1 + \alpha_2 z_2 + \alpha_3 z_3 + \alpha_4 z_4.$$

- Rank the four factors by the absolute values of their standardized coefficients.
- Which model is easier to use for comparing the relative importance of the four factors: the original model or the standardized model?
- Compare the fitted values from the original model and the standardized model. Are they the same?
- Explain your answer using the column-space view of regression.

Hint: Standardizing predictors changes the scale of the coefficients, but it does not change the information available to the model. With an intercept included, the original and standardized design matrices span the same fitted-value space.

Task 4: Quantify uncertainty. For your fitted Four Factors model, compute and interpret uncertainty.

- Report the residual standard error
$$\widehat{\sigma} = \sqrt{\frac{RSS}{n-p}},$$
where p is the number of fitted coefficients, including the intercept. Interpret $\widehat{\sigma}$ in the units of wins.
- Report the standard error and a 95% confidence interval for each coefficient.
- Which factors have estimated effects that are clearly distinguishable from zero? Which factors are more uncertain?
- Choose one team from the dataset. Using its four factor values, compute:
  (a) a point prediction for wins;
  (b) a 95% confidence interval for the team's expected wins given its four factor profile;
  (c) a 95% prediction interval for the team's actual wins.
- Which interval is wider: the confidence interval for expected wins or the prediction interval for actual wins? Explain why.

Hint: The confidence interval describes uncertainty in the fitted mean response. The prediction interval is wider because it also includes the randomness of an individual team's season outcome.

Task 5: Prediction. Use an out-of-sample prediction test to evaluate predictive performance.

- Randomly split the data into training and test sets.
- Fit the original and standardized models on the training data.
- Compute test-set RMSE for both models.
- Which model predicts better? Why?

2.2 Expected Outcome of a Punt

2.2.1 Data

We have a dataset consisting of punts, where each row represents a single punt. Each row contains the following variables:

- i: index of the i^{th} punt
- y_i: the outcome variable, which is the next yard line from the opponent's perspective after the punt
- ydl_i: the yard line from which punt i took place, measured in yards from the opponent's goal line
- pq_i: the punter quality of the punter who kicked punt i
- $punter_i$: the name of the punter who kicked punt i

The goal is to model expected post-punt field position and then use that model to evaluate punters relative to expectation.

2.2.2 Your Assignment

Task 1: Explore the data.

- Plot y_i against ydl_i.
- Bin punts by starting field position and plot the average post-punt yard line in each bin.
- Describe the shape of the relationship. Does it look linear? Where does it bend?
- Plot or summarize the distribution of punter quality pq_i.

Task 2: Fit competing models. Use multivariable regression to model the next yard line for the opponent as a function of current yard line and punter quality. At minimum, try the following models:

$$E[y_i \mid ydl_i] = \beta_0 + \beta_1 ydl_i,$$

$$E[y_i \mid ydl_i] = \beta_0 + \beta_1 ydl_i + \beta_2 ydl_i^2,$$

$$E[y_i \mid ydl_i, pq_i] = \beta_0 + \beta_1 ydl_i + \beta_2 ydl_i^2 + \gamma pq_i,$$

and a spline model such as

$$E[y_i \mid ydl_i, pq_i] = \beta_0 + \sum_{k=1}^{K} \beta_k B_k(ydl_i) + \gamma pq_i,$$

where $B_1, \ldots, B_K$ are spline basis functions.

- Visualize the fitted curves from each model.
- Use train/test RMSE or cross-validation to select a model.
- Explain the tradeoff between the linear, quadratic, and spline fits.
- Does adding punter quality improve out-of-sample prediction?
- Interpret the coefficient on punter quality, if your selected model includes it.

Task 3: Visualize uncertainty in the punt model. After selecting your preferred model, visualize both the fitted mean response and uncertainty around the fit.

- Plot the expected opponent yard line after the punt as a function of the starting yard line.
- Add a 95% confidence band for the expected opponent yard line.
- Add a 95% prediction band for the outcome of one individual punt.
- Explain why the prediction band is wider than the confidence band.

- Where is the model most uncertain? Does uncertainty increase in parts of the field with fewer punts?

Hint: In R, `predict(model, interval = "confidence")` *gives uncertainty for the expected response, while* `predict(model, interval = "prediction")` *gives uncertainty for one new observation.*

Task 4: Interpret punt yards over expected. Rank the punters by punt yards over expected (PYOE), but include uncertainty in the ranking.

First, define PYOE carefully, including the sign convention. Since larger realized post-punt yard line is better than expected in this setup, define

$$PYOE_i = y_i - \widehat{E}[y_i \mid x_i],$$

so that positive values correspond to better-than-expected punts.

- Compute $PYOE_i$ for each punt.
- For each punter, compute:
  (a) average PYOE;
  (b) number of punts;
  (c) standard error of average PYOE.
- Rank punters by average PYOE.
- Visualize the punter rankings with uncertainty intervals.
- Which punters appear clearly above average?
- Which rankings are unstable because of small sample size or high punt-to-punt variability?

Final reflection. Answer the following in a short paragraph.

1. How did adding columns to the design matrix change what the model could fit?
2. When did extra flexibility help?
3. When might extra flexibility hurt?
4. What does $\widehat{\sigma}$ tell you about the typical size of model errors?
5. Why is a prediction interval wider than a confidence interval for the expected response?
6. What is one coefficient, prediction, or punter ranking from this lab that you would be cautious about interpreting too strongly?